

Date

Applied Thermodynamics

DAY-1

The Bridge :-

Real World Application :-

Three forms of energy

- Thermal
- Kinetic
- Chemical

→ 1st law of TD is the ~~the~~ law of conservation of energy which is "energy can neither be created nor be destroyed."

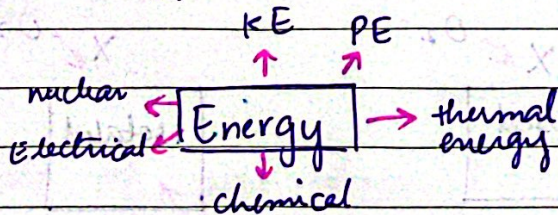
DAY-2

Introduction to thermodynamics :-

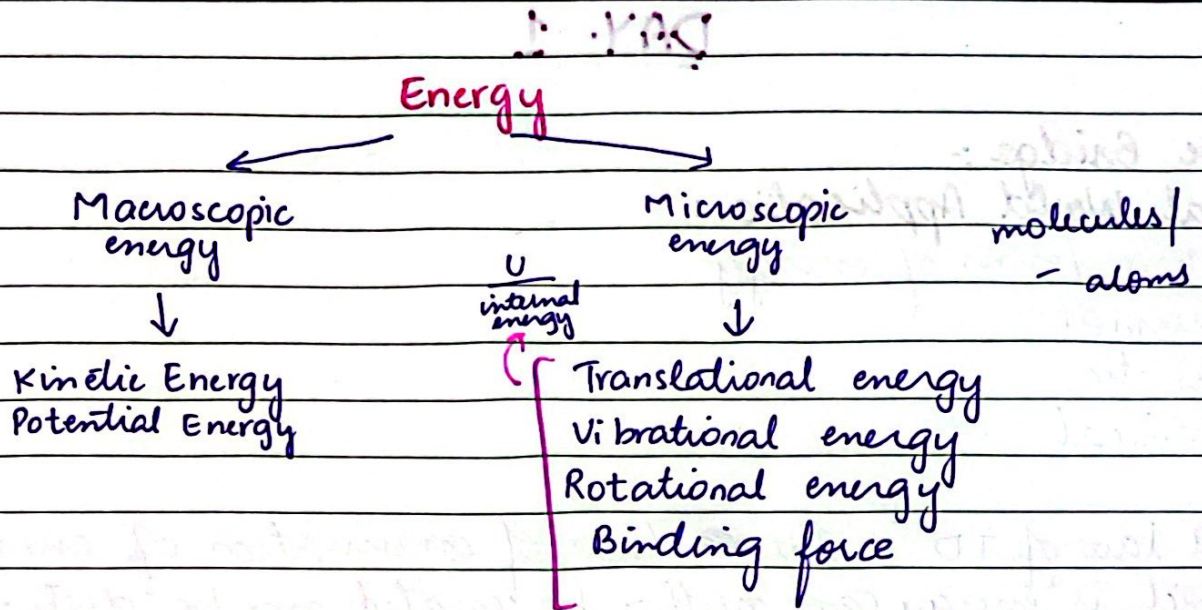
Deals with the transfer of energy from one form to another or from one place to another.

① Energy transfer ② heat, work and mass :-

- ① A quantitative property that must be transferred from a body or physical system to perform work or transfer heat.
- ② Form of energy that is transferred between two bodies due to difference in the temperature.



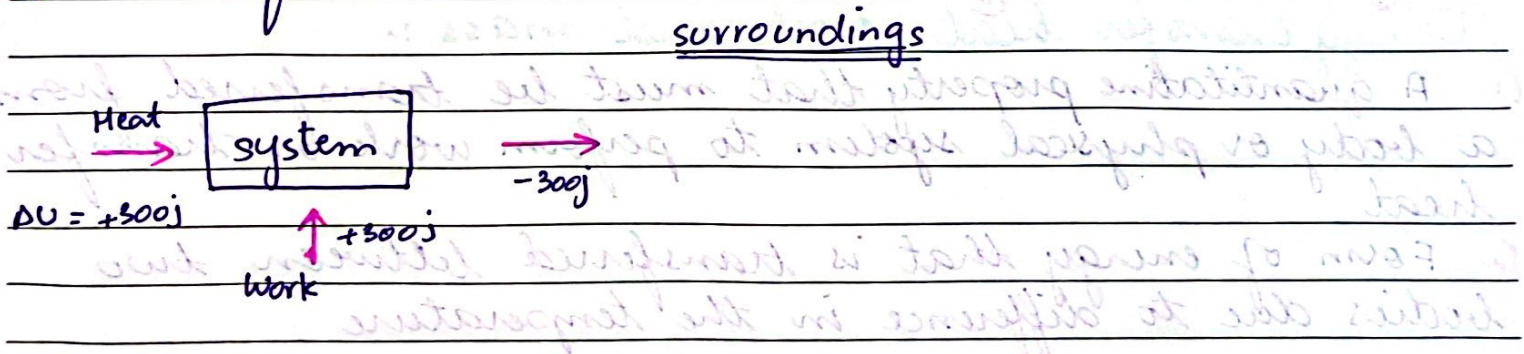
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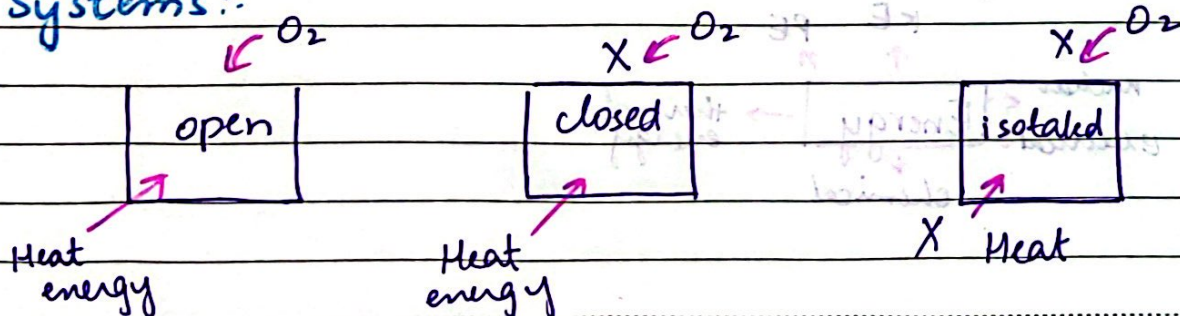
$$\text{Total energy} = KE + PE + U$$
$$= \frac{1}{2}mv^2 + \cancel{\text{energy}} U$$

Mechanisms of energy transfer :-

1. Work done (W) → Joule → Mechanical energy
2. Heat (q) → Joule → Thermal energy
3. Mass flow



Systems :-



Date

$$\Delta U = q + w \quad (\text{chem})$$

$$\Delta U = q - w \quad (\text{Physics})$$

chem

$w = -ve \rightarrow$ work done by the system

$w = +ve \rightarrow$ Work done on the system

physics

$w = -ve \rightarrow$ Work done on the system

$w = +ve \rightarrow$ Work done by the system

Heat:

$q \rightarrow +ve$ (absorb) $\rightarrow$ endothermic

$q \rightarrow -ve$ (release) $\rightarrow$ exothermic

Example:

100 J of work is done on the system. U increases by 74 J. How much energy is transferred as heat?

sol.

$$\Delta U = q - w$$

$$74 = q - (-100)$$

$$74 = q + 100$$

$$q = 74 - 100$$

$$q = -26 \text{ J}$$

Example:

A sample of gas does 150 J of work against its surrounding and loses 90 J of internal energy

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in the process Does the gas lose or gain the heat And How much?

Sol.

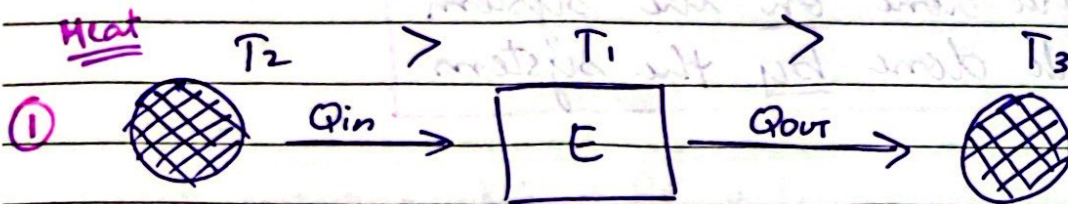
$$\Delta U = q - W$$

$$-90 = q - 150$$

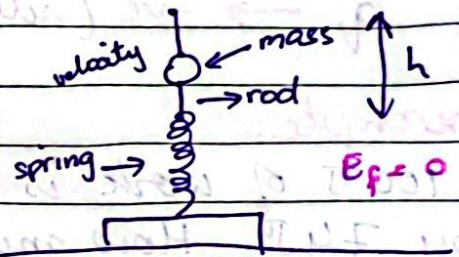
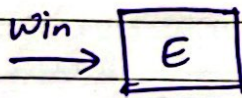
$$q = 150 - 90$$

$$q = 60 \text{ J} \quad (\text{gain}) \quad (\text{endothermic}) \quad (\text{absorb})$$

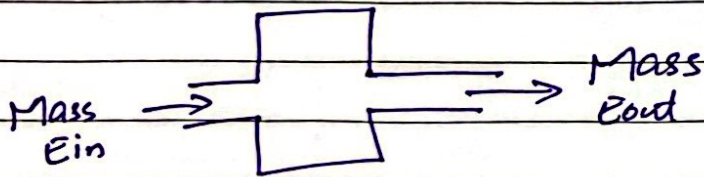
Heat



② Work



③ Mass



$$W - p = U \Delta$$

$$(100) - p = p \Gamma$$

$$100 + p = p \Gamma$$

$$100 - p \Gamma = p$$

$$100 - p \Gamma = p$$

in the process Does the gas lose or gain the heat And How much?

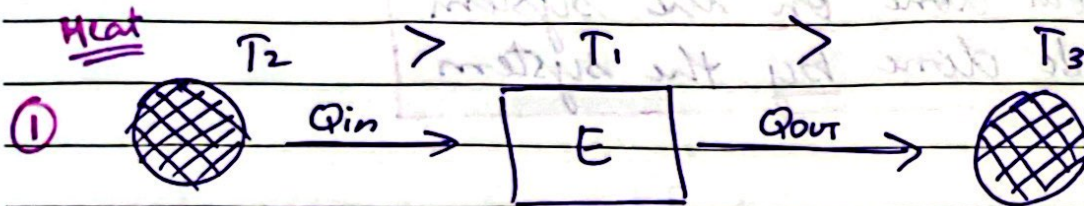
Sol.

$$\Delta U = q - W$$

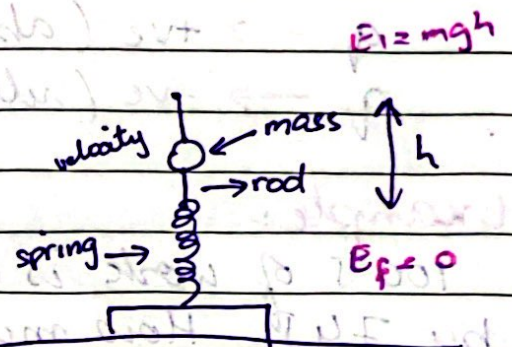
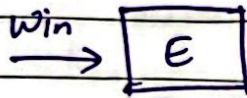
$$-90 = q - 150$$

$$q = 150 - 90$$

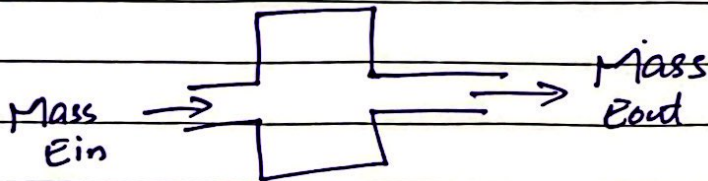
$$q = 60 \text{ J} \quad (\text{gain}) \quad (\text{endothermic}) \quad (\text{absorb})$$



② Work



③ Mass



DAY - 3

Properties of Pure Substance:

Pure substance:

A pure substance is defined as a material that has a homogeneous and invariable chemical composition.

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Homogeneity :-

The substance is the same throughout, with no regions having a different composition.

Invariable Composition :-

The chemical composition doesn't change even when the substance transitions between different phases.

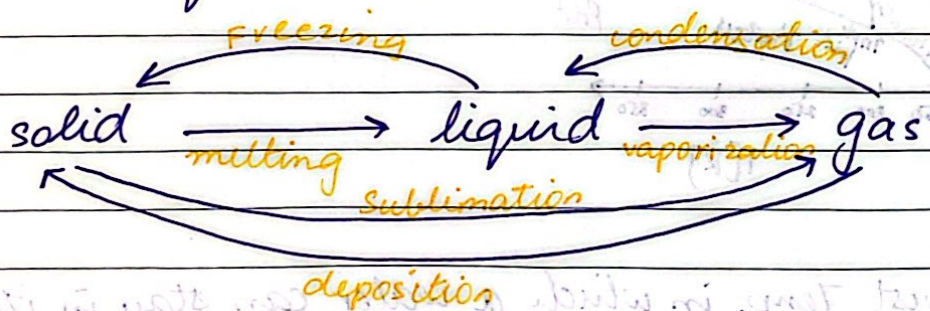
Solid :- Tightly packed molecules

Liquid :- Tightly packed but moveable

Gas :- Freely moveable

Phase Changes And Diagrams

Phase change of a material occurs when it transitions from one phase / state to another.

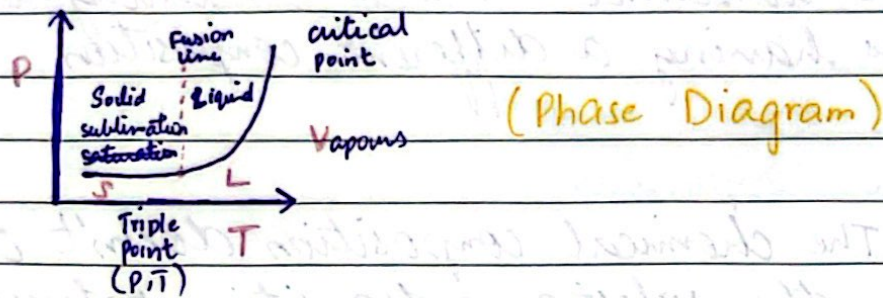


Saturation Temperature :-

The specific temp. at which a substance undergoes a phase change under a given

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pressure. (vaporization, condensation)

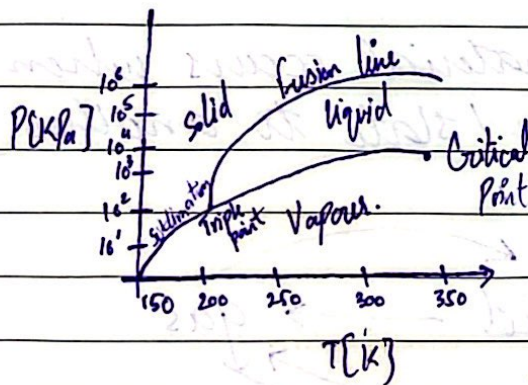


Triple point of water occurs at 0.01°C , 0.6117 kPa

Critical Point:

It marks the end of vapourization curve on a phase diagram. Beyond this point liquid and gaseous phase becomes indistinguishable.

Critical point of water: 374.14°C and 22.09 MPa



0.01°C is the lowest Temp. in which a water can stay in its lowest state.

QUESTION :- If the pressure is smaller than the smallest

Date

pressure P_{sat} at a given temperature T . What is the phase?

Date

pressure P_{sat} at a given temperature T . What is the phase?

Sol. On google classroom

DAY-4

The P-v-T surface:-

- Crucial for understanding how substances behave under different conditions (P, v, T)
- Applications $\rightarrow$ Refrigeration, power generation and compressor etc
- Captures the relationship b/w the pressure, volume and temperature

P-v-T surface of water:-

- Ice — has larger volume than liquid state
- liquid water

The fusion line (solid - liquid) line on pressure - temp diagram has a negative slope indicating the increasing the pressure decrease the freezing point

pressure $\uparrow$ boiling point $\downarrow$ freezing point $\uparrow$

P-v-T surface of a typical substance:

Shrinks on freezing

Solid phase have lower volume than liquid phase.

The fusion line (solid - liquid) & on pressure - temp

diagram has a positive slope indicating the increasing the freezing pressure increases the freezing point

Phase Diagram:-

P-T diagram

P-V diagram

- Fusion line
- Sublimation line
- Triple point
- Critical point

T-V Diagram:-

- Shows how volume changes with temperature at constant pressure illustrating phase change from liquid to vapours.
- Depicts the boiling phase

P-V Diagram:-

- Shows how pressure changes with specific volume at constant temperature

Independent Properties:-

The state of substance can be defined by two independent properties. However, in the two phase region (during boiling or condensation), pressure and temperature are independent because they are fixed at a saturated point for a given pressure-temperature pair.

DAY-5

Tables of Thermodynamics Properties :-

Purpose:- Thermodynamics tables provide essential data for various substance that describe their thermodynamic properties (like temperature, pressure, volume, specific volume, internal energy like enthalpy and entropy).

Saturation Temperature (T_{sat})

Temperature at which phase change occurs at const T

Saturation Pressure (P_{sat}) :-

Pressure at which phase change occurs at const T .

This pair is not a group of independent properties.

T	P	V_f	V_{fg}	V_g
0	P_1	V_1	V_{11}	V_{g1}
10	P_2	V_2	V_{12}	V_{g2}
20	P_3	V_3	V_{13}	V_{g3}
30	P_4	V_4	V_{14}	V_{g4}
40	P_5	V_5	V_{15}	V_{g5}

- Linear interpolation
- Polynomial regression
- Moving average
- Lagrange interpolation

Saturated liquid & vapour tables

2-VAP

Table B 1.2 :-

- v_f - specific volume of a saturated liquid
- v_g - specific volume of a saturated vapour
- v_{fg} - Difference b/w v_g & v_f

$$v_{fg} = v_g - v_f$$

First column :- temperature

Second column :- Corresponding saturation pressure (P_{sat})

Table B 1.2 :-

- contains the same information as in B1.1 but organized pressure
- Useful

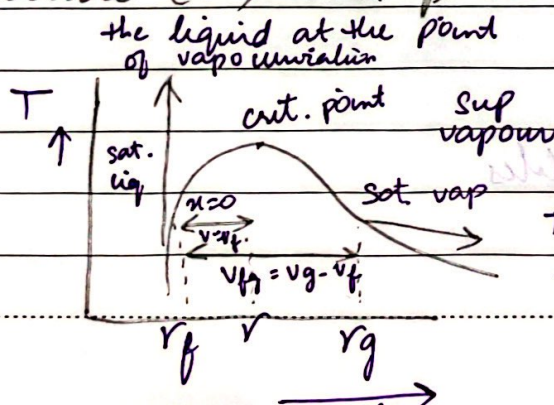
→ Sir was absent on Day 6.

DAY-6

The Two Phase states :-

Two phase state :-

A mixture of two phases of the same substance, such as liquid and vapour, in equilibrium at the same temperature (T) and pressure (P).



Date

Specific Volume:

Volume occupied by a unit mass of a substance measure of how much space 1kg of substance occupies.

$$v = \frac{V}{m} \Rightarrow V = v m$$

Specific Volume in Two-phase Mixture:-

Total volume of two-phase mixture is the sum of the volumes of the liquid (V_{liq}) and the vapour (V_{vap})

$$V_{Total} = V_{liq} + V_{vap} = m_{liq} v_f + m_{vap} v_g$$

Avg. specific volume:

quality
↑

$$v = \frac{V}{m} = \frac{m_{liq} v_f}{m} + \frac{m_{vap} v_g}{m}$$

$$\left(\frac{m_{liq}}{m} = (1-x), \frac{m_{vap}}{m} = x \right)$$

$$v = (1-x) v_f + x v_g$$

$$v = (1-x) v_f + x v_g = v_f - x v_f + x v_g \Rightarrow v = v_f + x(v_g - v_f)$$

$$v = v_f + x v_{fg}$$

Quality:-

It is the ratio of mass of vapours to the total mass in a two phase mixture.

Range: $0 \leq x \leq 1$

Problem:..

Calculate the specific volume of a saturated mixture of water at 200°C with a quality of 70%.

Refer: Table B1.1

$$\begin{aligned} V &= x V_g + V_f \\ &= x(V_g - V_f) + V_f \\ &= 0.7(0.12736 - 0.001156) + 0.001156 \\ &= 0.0894988 \text{ m}^3/\text{kg} \end{aligned}$$

Date

Range : $0 \leq x \leq 1$

Problem :-

calculate the specific volume of a saturated mixture of water at 200°C with a quality of 70%.

Refer : Table B1.1

Sol :-

$$\begin{aligned} V &= x V_{fg} + V_f \\ &= x (V_g - V_f) + V_f \\ &= 0.7 (0.12736 - 0.001156) + 0.001156 \\ &= 0.0894988 \text{ m}^3/\text{kg} \end{aligned}$$

DAY-7

Ideal Gas Law:-

Describes the behaviour of gases under certain conditions. It provides simple correlation among pressure, temperature (T) and specific volume (V). It is an approximation and works best under high temperature, low density and low pressure.

Ideal Gas Equation of state :-

$$PV = RT$$

For calculations involving mass / moles :-

$$PV = mRT = nRT$$

$$R = \frac{\bar{R}}{M}$$

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$$\bar{R} = 8.3145 \frac{\text{kJ}}{\text{K mol K}}$$

$$PV = mRT = n\bar{R}T$$

n is the number of moles

$$n = \frac{m}{M}$$

- Low pressure
- High temperature
- Low density

Mass flow rate — Ideal Gas Law

QUESTION :: (Table 8.5)

Mass of air?

Temp: 25°C , $P = 100 \text{ kPa}$, Molecular mass = 18.015 kg/mol

Sol: —

$$PV = mRT \quad (R = 0.2875 \text{ kJ/kg.K})$$

$$m = \frac{PV}{R\bar{T}}$$

$$m = \frac{100 \times 240}{0.287 \times 298}$$

$$0.287 \times 298$$

$$R = \frac{\bar{R}}{M} = \frac{8.3145}{28.97} = 0.287$$

$$\bar{R} = 8.3145$$

QUESTION:-

Tank has a volume = 0.5 m^3

Date

$m = 10 \text{ kg}$ for ideal gas.
molecular mass = 24, $T = 25^\circ \text{C}$

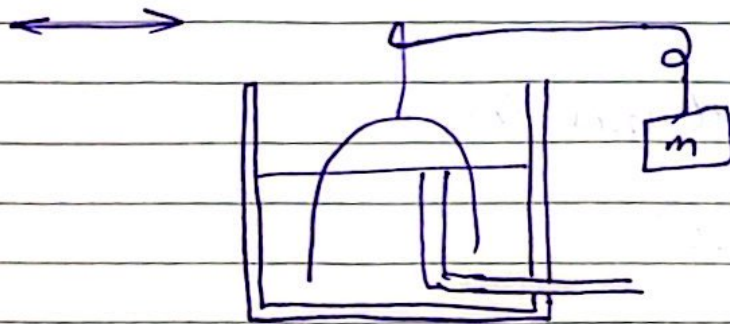
sol

$$PV = mRT$$

$$P = \frac{mRT}{V}$$

$$= \frac{10 \times 0.3456 \times 298}{0.5}$$

$$= 2064 \text{ kPa}$$



A gas bell is submerged with its mass counterbalanced with rope and pulleys. The pressure is 105 kPa and the T is 21°C . Volume increase is measured to be 0.75 m^3 over a period of 185 seconds. What is the volume flow rate and mass flow rate.

$$R = 0.1889 \text{ kJ/kg} \cdot \text{K}$$

$$PV = mRT$$

$$\dot{V} = \frac{0.75 \text{ m}^3}{185 \text{ s}} = \frac{\Delta V}{\Delta T}$$
$$= 0.00405 \text{ m}^3/\text{s}$$

Date

$$P\dot{V} = \dot{m}RT$$

$$\dot{m} = \frac{P\dot{V}}{RT}$$

$$= \frac{(105 \text{ k})(0.00405)}{(0.1889 \text{ k})(294)}$$

$$= \cancel{7.6 \text{ m}}$$

$$= 0.00766 \text{ kg/s}$$

DAY-8

Compressibility Factor (z):

To account for the deviations in ideal gas law, compressibility factor is used

$$Z = \frac{PV}{RT}$$

$$PV = ZRT$$

($Z = 1 \rightarrow$ Gas is ideal)

($Z \neq 1 \rightarrow$ Non-ideal behaviour)

If $Z < 1 \rightarrow$ attractive forces b/w molecules dominate causing the gas to be more compressible.

$Z > 1 \rightarrow$ Repulsive forces dominate $\rightarrow$ gases resist compression.

$\rightarrow$ At low pressures, Z approaches 1 as pressure approaches 0.

$\rightarrow$ At high temperatures, for many gases like nitrogen and air, Z remains close to 1 upto pressures of around 10 MPa.

This means ideal gas law is still a good approximation even at moderate pressures.

Date

Reduced Pressure:-

$$P_r = \frac{P}{P_c}$$

(P_c : critical pressure)

Reduced temp:-

$$T_r = \frac{T}{T_c}$$

(T_c : critical temp)

These reduced properties allows us to plot z as function

8-VAD

Problem 6

Specific volume v at 200°C with the quality of 70%.

$$v = x v_{fg} + v_f = x (v_g - v_f) + v_f = 0.089488 \text{ m}^3/\text{kg}$$

∴ By table B.1.1

Compressed Liquid

A liquid state at a pressure higher than the saturation pressure for a given temp. OR at a temp. lower than the saturation temp for given pressure.

$$\begin{aligned} P &> P_{\text{sat}} \text{ at a given } T \\ T &< T_{\text{sat}} \text{ at a given } P \\ v &< v_f \text{ at a given } T \text{ or } P \end{aligned}$$

$$\begin{aligned} v &< v_f \rightarrow \text{Internal Energy} \\ h &< h_f \rightarrow \text{Enthalpy} \end{aligned}$$

Superheated Vapors

A liquid state at a pressure lower than the saturation pressure OR at a temp higher than saturation temp

$$\begin{aligned} P &< P_{\text{sat}} \text{ at given } T \\ T &> T_{\text{sat}} \text{ at given } P \\ v &> v_g \text{ at given } T \text{ or } P \end{aligned}$$

$$\begin{aligned} v &> v_g \rightarrow \text{Internal energy} \\ h &> h_g \rightarrow \text{Enthalpy} \end{aligned}$$

Date

Reduced Pressure:-

$$P_r = \frac{P}{P_c}$$

(P_c : initial pressure)

Reduced temp:-

$$T_r = \frac{T}{T_c}$$

(T_c : initial temp)

These reduced properties allows us to plot z as function

(After Mid 1)

DAY 9 8-VAD Chapter 3

First law of Thermodynamics For controlled Mass:-

(law of conservation of energy)

Control Mass:- A fixed mass system mass doesn't change.

Control volume:- A spatial region where mass can enter and exit.

Heat:- Energy transfer due to temp. difference

Work Transfer:- Energy transfer due to the force applied over a distance.

Energy Equation For controlled Mass:-

Instantaneous Rate Flow:-

$$E_{cv} = \dot{Q} - \dot{W}$$

$$\frac{dE_{cv}}{dt} = \dot{Q} - \dot{W}$$

• $\dot{E}$: rate of change of total energy within the control mass.

Date

Sign Convention :-

Heat (Q) : (+ve) $\rightarrow$ when heat is entering the system

W (W) : (+ve) $\rightarrow$ Work done by the system

Energy Equation For Finite changes :-

Consider total energy changes over a process from state 1 $\rightarrow$ state 2

Starting Point :-

$$dE_{cv} = \delta Q - \delta W$$

Integration over time

$$\int_{t_1}^{t_2} dE_{cv} = \int_{t_1}^{t_2} \delta Q - \int_{t_1}^{t_2} \delta W$$

$$E_2 - E_1 = Q_{1-2} - W_{1-2}$$

$\rightarrow$ Total heat added
From state 1 $\rightarrow$ state 2

Energy Equation (Detailed) :-

$$E_{Total} = U + K.E + P.E$$

$$E_2 - E_1 = (U_2 - U_1) + \frac{1}{2} m(V_2^2 - V_1^2) + mg(h_2 - h_1)$$

QUESTION :-

$$Q = +500 \text{ J}$$

$$W = +200 \text{ J}$$

$$U = 500 - 200$$

$$U = 300 \text{ J}$$

Date

State Functions

Depend only on the state of system e.g. H, E

Path Functions

Depend on the specific process path taken!
eg: Q, w

Pressure Volume Work

$$\delta W = P dV = \frac{F}{dA} dA \cdot ds$$

$$\delta W = F \cdot ds$$

Isothermal: $T \rightarrow$ same

$$PV = nRT$$

Isothermal equation of an ideal gas in term of work $\rightarrow$ homework

$\rightarrow$ Isothermal Process

Constant Temperature

$$PV = nRT$$

$\therefore$ equation in term of work (homework)

$\rightarrow$ Isobaric Process:

Constant Pressure

$$w = P(V_2 - V_1)$$

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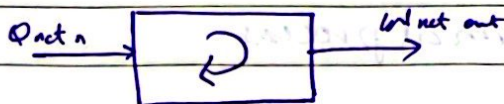
MAHE/UP/ LECTURE DAY - 10

First Law of thermodynamics - Control Mass & Cyclic processes

$$0 = \oint \delta Q - \oint \delta W$$

It implies that there is no net change in the total energy of control mass and the rate of change of energy is zero.

$$\text{cyclic integral of heat transfer} \leftarrow \oint \delta Q = \oint \delta W \rightarrow \text{cyclic integral of the work done}$$



QUESTION :-

A engine is absorbing 250 kJ of heat and releasing 150 kJ of energy. Find First law of thermodynamics

SOL :-

$$\eta = \frac{150}{250} \times 100 = 60\%$$

$$\left(\eta = \frac{W_{\text{net out}}}{Q_{\text{net in}}} \right)$$

Question :-

$$P_1 = 300 \text{ kPa}, V_1 = 0.05 \text{ m}^3$$

the gas undergoes an isothermal expansion until its volume doubles. During expansion, the gas absorbs 100 J of heat from surroundings.

① $P_2 = ?$

② Calculate work done by the gas during expansion

③ What is the internal energy change (ΔU)?

Date

MAKESOP/RECURSE DAY-10

$$P_1 V_1 = P_2 V_2$$

$$P_2 = 150 \text{ KPa}$$

Work done by the ideal gas

$$W = n R T \ln\left(\frac{V_2}{V_1}\right)$$

$$W = P_1 V_1 \ln\left(\frac{V_2}{V_1}\right)$$

$$W = 10.4 \text{ KPa}$$

For ideal gas under isothermal process

$$\Delta U = 0$$

$$\Delta U = 0 - W \Rightarrow \Delta U = -W$$

QUESTION :-

$$\text{Mass} = 1100 \text{ Kg}$$

$$\text{K.E} = 400. \text{KJ}$$

$$V = ?$$

$$\text{KE} = \frac{1}{2} m v^2$$

$$v = \sqrt{\frac{2 \text{KE}}{m}}$$

$$v = 26.96 \text{ m/s}$$

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height ?

$$P.E = K.E = nqh = \frac{1}{2} mv^2$$

$$h = \frac{PE}{mg}$$

$$= \frac{400000}{(100)(9.8)}$$

$$h = 37.106 \text{ m}$$

Question :-

$$m = 10 \text{ kg}$$

$$\text{liq water} = 100 \text{ kg}$$

$$h = 10.2 \text{ m}$$

same Temp (stage 1)

stone falls into water -

Determine

$$\Delta U, \Delta K.E, \Delta P.E, Q, W?$$

enter in water (stage 2)

Just come to rest in bucket (stage 3)

heat transferred to the surroundings that stone & water have same temp (last stage)

sol:-

$$① \Delta U = 0 \quad (\text{Temp const isothermal process}) \quad (\text{stage 1-2})$$

$$② Q_{1-2} = 0$$

$$③ W_{1-2} = 0$$

Date

$$\Delta KE + \Delta PE = 0$$

$$\Delta KE = -\Delta PE$$

$$= -mg(h_2 - h_1)$$

$$= 1 \text{ kJ}$$

stage 2 $\rightarrow$ 3

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Delta PE = 0, Q = 0, W = 0$$

$$\Delta U + \Delta KE = 0$$

$$\Delta U = -\Delta KE = -1 \text{ kJ}$$

stage 3 $\rightarrow$ 4

$$\Delta KE = 0$$

$$\Delta PE = 0$$

$$W = 0$$

$$\Delta U = -1 \text{ kJ}$$

$$Q = -1 \text{ kJ}$$

DAY-11

** Modes Of Heat Transfer ::

Heat :: Energy transferred across the boundary of the system from one body to another (surroundings) due to temperature difference b/w two systems. Heat naturally flows from a system at high temp. to one at a lower temperature. If heat doesn't flow $\rightarrow$ thermal equilibrium

units ::

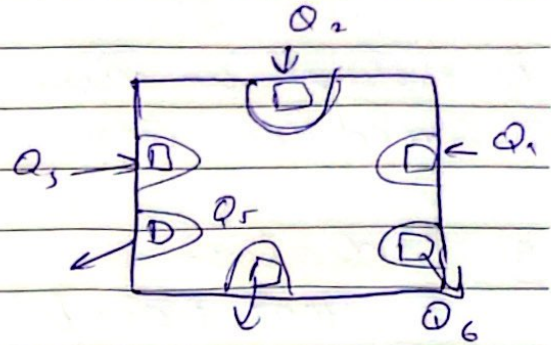
Joule (SI) BTU, Calories

sign convention : Heat into the system : +ve
Heat out of the system : -ve

Date

Path function

$$Q_{\text{total}} = \int \delta Q$$



① Conduction :-

Transfer of heat due to the direct contact b/w particles of matter. It happens because higher energy particle (those at higher temp) transfer energy to lower energy molecules (those at lower temp) via molecular collisions.

Fourier's Law of Conduction :-

$$\dot{Q} = -KA \frac{dT}{dx}$$

$$K (\text{for metals}) = 100 \text{ W/m} \cdot \text{K}$$

$$K (\text{For isolators}) = \text{upto } 0.01 \text{ W/m} \cdot \text{K}$$

where

$\dot{Q}$ = rate of heat transfer

K = thermal conductivity of material

A = cross sectional area through which heat is transferred

dT/dx = Temperature gradient

② Convection :-

Heat transfer through a moving fluid (liq or gas). It is a combination of bulk motion of fluid and conduction.

Newton's law of cooling :-

$$\dot{Q} = hA \Delta T$$

$\nwarrow$
convective heat
transfer co-efficient

- ① Natural convection
- ② Forced convection

③ Radiation ::

Heat transfer due to electromagnetic waves and doesn't require any medium.

e.g Heat from sun that reaches Earth.

Stefan - Boltzman Law

$$\dot{Q} = \epsilon \sigma A T_s^4$$

where

ϵ = Emissivity of the surface (0 to 1)

σ = Stefan - Boltzman constant (5.67×10^{-8})

A = surface area

T_s = surface temp

Problem ::

Room temp = $T_{\text{room}} = 20^\circ\text{C}$

Outside temp = $T_{\text{ambient}} = -10^\circ\text{C}$

outer glass surface temp : $T_{\text{out}} = 12.1^\circ\text{C}$

Glass thickness : $x = 0.005\text{ m}$

$k = 1.4\text{ W/m}\cdot\text{K}$

$A = 0.5\text{ m}^2$

$h = 100\text{ W/m}^2\cdot\text{K}$

Conduction through glass?

convection from glass to ambient air?

Date

Sol :-

$$\dot{Q}_{\text{conduction}} = -kA \frac{\Delta T}{\Delta x}$$

$$\Delta T = 20 - 12.1 = 7.9^\circ\text{C}$$

$$\Delta x = 0.005 \text{ m}$$

putting the values

$$\dot{Q}_{\text{conduction}} = -1106 \text{ W}$$

$$\dot{Q}_{\text{convection}} = h A \Delta T$$

$$\Delta T = 12.1^\circ\text{C} - (-10^\circ\text{C})$$

$$= 22.1^\circ\text{C}$$

putting the values

$$\dot{Q}_{\text{convection}} = 1105 \text{ W}$$

Extensive vs Intensive Properties :

Extensive properties $\rightarrow$ depends on the amount of matter
(mass, volume, internal energy)

Intensive properties $\rightarrow$ independent of the mass of the
system (temp, pressure)

Internal Energy :-

$$\text{total internal energy} = \text{mass} \times \text{specific internal energy}$$

$U = m u$

$$V_{fg} = V_g - V_f$$

Date

Internal energy in the liquid-vapour region

$$U = U_{liq} + U_{vap}$$

$x=0$ (saturated liq)

$$v = (1-x)v_f + xv_g$$

$$u = v_f + x(v_g - v_f)$$

$x=1$ (saturated vap)